



Cloud Security with AWS IAM



yakubu abdullahi yusuf

Policy editor

visu

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": "ec2:*",  
7       "Resource": "*",  
8       "Condition": {  
9         "StringEquals": {  
10          "ec2:ResourceTag/Env": "development"  
11        }  
12      }  
13    },  
14    {  
15      "Effect": "Allow",  
16      "Action": "ec2:Describe*",  
17      "Resource": "*"   
18    },  
19    {  
20      "Effect": "Deny",  
21      "Action": [  
22        "ec2:DeleteTags",  
23        "ec2:CreateTags"  
24      ],  
25      "Resource": "*"   
26    }  
27  ]  
}
```



Introducing Today's Project!

In this project, I will demonstrate how to use IAM service to control who is authenticated and authorized in my AWS account. I'm doing this project to learn cloud security with AWS IAM.

Tools and concepts

Services I used were EC2 instances, IAM Policies, IAM Users and User Groups, AWS Account Alias and JSON policy generator. I also learnt some Key concepts which are IAM users and Groups, AMI's, instances and also Instances types.

Project reflection

This project took me approximately 30 minutes to complete. Another fun project to work on as someone stepping into cloud computing. The most challenging part of the project was not starting it earlier, I enjoyed it.



Tags

Tags are user-defined labels given to AWS resources for easier identification, organization and management.

The tag I've used on my EC2 instances is called Env The value I've assigned for my instances are production and development.

Key	Value
Env	production



IAM Policies

IAM Policies are rules that allows user to perform certain operations on or with the AWS resources.

The policy I set up

For this project, I've set up a policy using JSON

I've created a policy that allows full EC2 access only to resources tagged with Env=development while still allowing the user to describe all EC2 resources.

When creating a JSON policy, you have to define its Effect, Action and Resource.

The Effect specifies whether access is allowed or denied, action specifies which AWS operations are permitted, and resource identifies which AWS those actions apply to.



My JSON Policy

Policy editor visu

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "ec2:*",
7       "Resource": "*",
8       "Condition": {
9         "StringEquals": {
10          "ec2:ResourceTag/Env": "development"
11        }
12      }
13    },
14    {
15      "Effect": "Allow",
16      "Action": "ec2:Describe*",
17      "Resource": "*"
18    },
19    {
20      "Effect": "Deny",
21      "Action": [
22        "ec2:DeleteTags",
23        "ec2:CreateTags"
24      ],
25      "Resource": "*"
26    }
27  ]
}
```



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NextWork Student

nextwork.org

Account Alias

An account alias is a user-defined friendly name for AWS user account that can be used instead of account ID.

Creating an account alias took me less than a minute. Now, my new AWS console sign-in URL is `https://nextwork-alias-abdou.signin.aws.amazon.com/console`



✓ Alias for this account updated to nextwork-alias-abdou.



IAM Users and User Groups

Users

IAM users are individual accounts in AWS owned by people or applications, each with their own login credentials and permissions to access specific AWS resources safely.

User Groups

IAM user groups are collections of IAM users, which allows the managing of permissions for all the users in a group at the same time through the attaching of policies to the group.

I attached the policy I created to this user group, which means i can manage the permissions for a group of users faster and easier.



Logging in as an IAM User

The first way is Email and the second way is by downloading the CSV file.

Once I logged in as my IAM user, I noticed that the user has been denied access to some resources. This was because the IAM user is not the root user and there are certain policies applied to its IAM group earlier.

The screenshot shows the AWS IAM console interface during the 'Retrieve password' step of creating a new IAM user. At the top, a green banner indicates 'User created successfully' and provides a 'View user' button. Below this, a progress bar shows four steps: 'Specify user details', 'Set permissions', 'Review and create', and 'Retrieve password' (which is the current step). The main content area is titled 'Retrieve password' and includes a warning that this is the only time the password can be viewed. It contains a 'Console sign-in details' box with the following information:

- Console sign-in URL:** <https://nextwork-alias-abdou.signin.aws.amazon.com/console>
- User name:** nextwork-dev-abdou
- Console password:** A masked password with a 'Show' button to reveal it.

There is also an 'Email sign-in instructions' button with an external link icon.

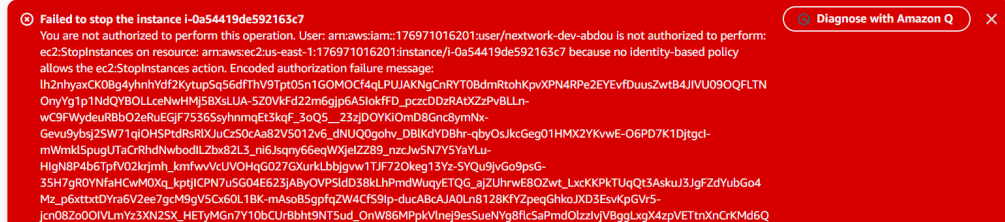


Testing IAM Policies

I tested my JSON IAM policy by logging into an IAM user account, tried to stop a production instance I applied the policy on, but received an error.

Stopping the production instance

When I tried to stop the production instance, a banner with an error message popped up. This was because the IAM user doesn't have the authorization to stop the production instance.

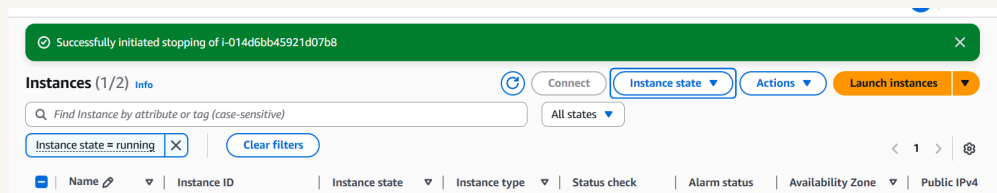




Testing IAM Policies

Stopping the development instance

Next, when I tried to stop the development instance, it stopped right away. This was because no policy was applied on the development instance.





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